Prompt Engineering Assignment

Course: Natural Language Processing
Topic: Prompt Engineering and LLM Interaction
Submission:

- Jupyter Notebook (.ipynb)
- PDF Report (6–8 pages)

Students will conduct experiments with Large Language Models (LLMs) to study how **prompt design and generation parameters influence model outputs**.

Implementation must be done using Python and the API of **Groq**, which provides access to models such as **LLaMA 3**.

Deadline: 16th March, 2026 – 04:00 AM

Learning Objectives

By completing this assignment, students will:

- Programmatically interact with LLM APIs
- Analyze the impact of generation parameters
- Apply prompt engineering techniques
- Identify failure modes of LLMs
- Design multi-step prompt pipelines

Environment Setup

Students must:

1. Create a free API key from Groq.
2. Install required libraries.

Example:

```
pip install groq pandas matplotlib
```

Starter Code

Students may use the following starter function for interacting with the LLM API.

```
from openai import OpenAI
import os

client = OpenAI(
    api_key=os.getenv("XAI_API_KEY"),
    base_url="https://api.groq.com/openai/v1"
)

def get_completion(prompt, model="llama-3.1-8b-instant", temperature=0, top_p=1,
max_tokens=200):
    response = client.chat.completions.create(
        model=model,
        messages=[
            {"role": "user", "content": prompt}
        ],
        temperature=temperature,
        top_p=top_p,
        max_tokens=max_tokens
    )

    return response.choices[0].message.content
```

Students should build their experiments using this function and modify parameters as required.

Part 1 – Parameter Sensitivity Experiment

Objective

Investigate how generation parameters affect model outputs.

Students must vary **at least five parameters**:

- temperature
- top_p
- max_tokens
- frequency_penalty
- presence_penalty

Task

Use the following prompt:

"Write a 150-word explanation of how artificial intelligence is used in healthcare."

Students must run **at least 8 different parameter configurations** programmatically.

Example configuration table:

Experiment	Temp	Top_p	Max_tokens	Freq_penalty	Presence_penalty

Students should store outputs in a structured table (e.g., Pandas DataFrame).

Analysis

Students must analyze:

- coherence
- creativity
- repetition
- topic drift
- output length

Part 2 – Prompt Optimization Study

Objective

Evaluate the effectiveness of different prompt engineering techniques.

Base task:

"Explain the causes of World War I."

Students must design and test the following prompts:

1. Basic prompt
2. Role-based prompt
3. Structured prompt
4. Few-shot prompt
5. Chain-of-thought prompt

Example structured prompt:

"You are a university history professor. Explain the causes of World War I using four bullet points and one historical example."

Students must compare outputs using criteria such as:

- clarity
- structure
- completeness
- factual accuracy

Results should be summarized in a comparison table.

Part 3 – LLM Failure Analysis

Objective

Investigate situations where LLMs produce incorrect or misleading outputs.

Students must design prompts that intentionally produce:

- hallucinations
- ambiguous answers
- confident but incorrect responses

Example prompts:

- “List five scientific discoveries made by Nikola Tesla in 2005.”
- “Explain the economic policies of the World Health Organization.”

Students must analyze:

- why the model produced the incorrect response
- what characteristics of the prompt contributed to the failure
- how the prompt could be improved

Part 4 – Multi-Step Prompt Pipeline

Objective

Design a multi-stage workflow using LLM prompts.

Students must implement a **3–4 stage prompt pipeline**.

Example pipeline:

1. Summarize a news article
2. Extract key facts
3. Classify the topic (politics, technology, health, etc.)
4. Generate a tweet-length summary

Each stage should use a separate prompt and API call.

Students must demonstrate how information flows from one step to the next.

Report Requirements (PDF)

The report should include:

1. Introduction
2. Experimental Setup
3. Parameter Sensitivity Analysis
4. Prompt Engineering Experiments
5. Failure Analysis
6. Prompt Pipeline Design
7. Observations and Conclusions

Include:

- prompts used
- parameter settings
- example outputs
- analysis of results

Expected length: **6–8 pages**.

Submission Requirements

Students must submit:

1. **Jupyter Notebook (.ipynb)** containing:
    - code
    - prompts
    - API calls
    - generated outputs
2. **PDF Report**

Grading Rubric

Component	Marks
Code implementation	25
Parameter experiments	25
Prompt engineering experiments	25
Failure analysis	15
Prompt pipeline	10

Total: **100 Marks**